Pal 2021 Breast — analysis summary

RUN 1 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Mixing + continuity judged on the regenerated embedding. iLISI 1.0 → 4.96 of 22 samples (Harmony θ=10); one continuous connected component (11 speck cells removed). A proliferation compartment (76% cycling, 14,165 cells) is resolved before the GO → the trajectory is scoped to the low-cycling developmental Body (45,191 cells).

Gate 2 — module geography → GO (non-branching)

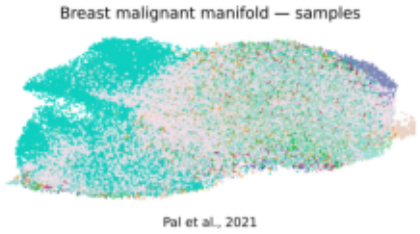
No Gast/pSom two-arm hub; a single embryonic axis runs pSom-high (origin) → Crest-high (terminus). Smoothed pSom vs Crest r=−0.93; joint pSom&Crest enrichment 0.23 (<1, inverse gradient). Crest terminal (high-decile pseudotime 0.92); Gast weak and flat.

Gate 3 — hub cross-repression → NA

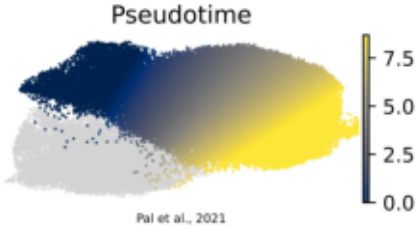
Branching-only gate → not applicable (single linear axis, no bifurcation to test for cross-repression).

STEP FIGURES

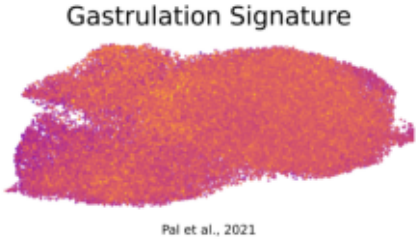
Manifold — samples



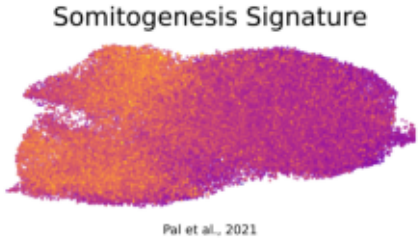
Pseudotime



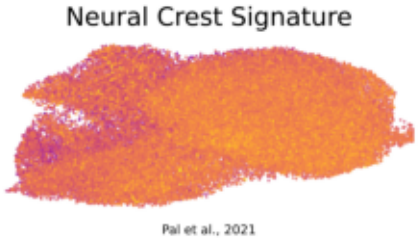
Gastrulation



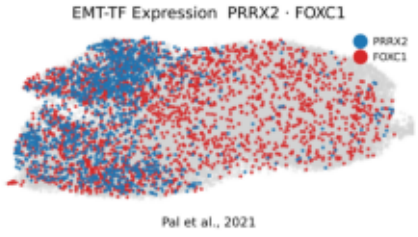
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast cancer (Pal et al., 2021), 10x — 45 samples / 305,224 cells (master scope); 59,356 malignant cells (22 samples). Trajectory scoped to the low-cycling developmental Body (45,191 cells; 14,165-cell cycling compartment excluded).

Verdict.
The embryonic pattern reproduces as the NON-BRANCHING case — one continuous, sample-integrated malignant manifold carrying a single somitogenesis→neural-crest developmental axis, with no Gast/pSom bifurcation. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs PRRX2 (strongest root) & FOXC1 (most expressed, 16%) — all detected factors origin-enriched (no terminus factor). Module strength Gast μ=−0.007 · pSom μ=+0.036 · Crest μ=−0.034 → Linear-pSom/Crest.

Parameter choices

Min-cell threshold	≥12 (geom-mean of 56→3 cliff, ratio 18.67); keeps 22 of 26	**Integration**	Harmony on sample, θ=10 (swept 2/5/10); n_pcs=20, spread=1.3, 3733 overdispersed
Sub-region scoping	Leiden res=0.15 → 2 lobes; Cycling lobe (76% cycling) excluded	**Geometry**	Non-branching curve (scFates N=20 λ=0.3); 2 tips, 0 forks, fold-free (SA 2.98)
Root node	curve tip 0 — pSom/Gast-high, Crest-low, 9% cycling; pt vs prolif ρ=+0.01	**EMT-TFs**	PRRX2 (strongest root) & FOXC1 (16%) — all detected origin-enriched; VIM excluded